

Abhinav Singh

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SUMMARY

AI/ML Engineer with 3+ years of experience delivering high-impact machine learning solutions in healthcare and pharmaceutical analytics. Expertise in LLM fine-tuning, statistical modeling (Clustering, Random Forest), and scalable data engineering on Databricks. Proven track record of automating complex workflows and translating business needs into deployable, ROI-driven ML systems.

EXPERIENCE

Associate | *Axtria*

May 2024 – Present

- Engineered ML-driven response curve models to replace legacy rule-based systems, increasing targeting precision by **35%** for pharmaceutical outreach.
- Developed an **LLM-powered RAG pipeline** for automated query resolution, reducing response latency by **90%** through fine-tuned models.
- Architected a scalable data system on **Databricks** for 500K+ physician records, improving processing speed by **50%** and reducing storage costs by **40%**.
- Applied K-Prototypes clustering and Random Forest to segment healthcare providers based on complex multi-dimensional datasets.

Analyst | *Axtria*

July 2022 – May 2024

- Automated end-to-end call planning by migrating Excel workflows to SQL, reducing execution time from **8 hours to 45 minutes** with 95% data quality improvement.
- Implemented **Greedy Optimization algorithms** to maximize physician coverage, resulting in a **25% increase** in outreach efficiency.
- Spearheaded cross-functional teams to manage high-stakes client deliverables, ensuring error-free deployment of targeting strategies.

SKILLS

Languages: Python (Scikit-learn, PyTorch), SQL, Java

Machine Learning: LLMs (Fine-tuning, RAG), Natural Language Processing, Deep Learning, Statistical Modeling, Optimization Algorithms

Tools & Cloud: Databricks, Apache Spark, Pandas, NumPy, Git, MS Excel

Software Engineering: OOPs, Data Structures & Algorithms, CI/CD Pipelines

PROJECTS

LLM-Based BQC Automated Response System

- Built an NLP tool using a fine-tuned LLM to interpret and answer natural language queries from field representatives in real-time.
- Designed a retrieval-augmented generation (RAG) pipeline to ensure high accuracy in technical query resolution.

Response Curve Modeling & Segmentation

- Optimized marketing ROI by modeling response curves on large-scale pharma datasets using K-Prototypes and Random Forest.
- Delivered a profit-maximizing targeting framework currently used for operational decision-making.

EDUCATION

Indian Institute of Information Technology (IIIT)

Allahabad, UP

M.Tech in Software Engineering (GPA: 9.1/10)

2020 – 2022

Madan Mohan Malaviya University of Technology

Gorakhpur, UP

B.Tech in Computer Science Engineering (GPA: 7.1/10)

2016 – 2020